

Pre-semester Reading Plan

Social Data Science specialization — MSc Economics, AAU

Autumn 2026 (modules: Data Handling & Applied ML, NLP & Network Analysis, Semester Project)

Plan window: 19 May 2026 – 31 August 2026 (15 weeks)

Overview

Fifteen-week ramp from thesis-defence wind-down to semester start. Weeks 1–4 are light (1–2 evenings/week, ~3 h); weeks 5–15 ramp to 8–10 h/week. Free items are marked with a dagger (†). Each phase has a concrete deliverable so progress is verifiable. Project framing: **Health Economics with SHARE** (Survey on Health, Ageing and Retirement in Europe).

Books already on the shelf (used throughout, primary in Phases 1–2):

- Rogel-Salazar, J. — *Data Science and Analytics with Python* (CRC, 2017). [RS-1]
- Rogel-Salazar, J. — *Advanced Data Science and Analytics with Python* (CRC, 2020). [RS-2]

1. Phase 0 — Decompression & orientation (weeks 1–2, 19 May – 1 Jun)

Goal: pick a project topic, install the toolchain.

- Skim Börsch-Supan (2013), *Data Resource Profile: SHARE*, IJE 42(4)[†], plus *SHARE Wave 9 Methodology* (Ch. 1–3)[†].
- Browse the *SHARE Compendium of Research*[†] TOC; bookmark 5–10 papers near your interest.
- Set up a conda env: `pandas`, `scikit-learn`, `statsmodels`, `linearmodels`, `networkx`, `nlTK`, `spacy`, `gensim`, `matplotlib`, `seaborn`, `jupyterlab`.
- **Deliverable:** 1-page note with 3 candidate SHARE project questions.

2. Phase 1 — ML foundations refresh (weeks 3–5, 2 – 22 Jun)

Goal: ground the Python/ML basics; use RS-1 as the spine and ISLP for the statistical theory.

- **RS-1 Ch. 1–3** — Python/NumPy/pandas refresher, data wrangling, exploratory analysis.
- **RS-1 Ch. 4–5** — statistics + linear regression in Python; pair with **ISLP Ch. 2–3**[†] for the theory.
- **RS-1 Ch. 6** (classification) alongside **ISLP Ch. 4**[†] and **Géron Ch. 1–3** — replicate one end-to-end pipeline on a SHARE subset.
- **VanderPlas Ch. 3 (pandas) + Ch. 4 (matplotlib)**[†] as reference only.
- **Deliverable:** notebook that loads easySHARE, cleans it, and runs logistic regression on a binary outcome with train/test split.

3. Phase 2 — Supervised & unsupervised ML depth (weeks 6–8, 23 Jun – 13 Jul)

Switch the spine to RS-2 (advanced) once RS-1 is finished.

- **RS-1 Ch. 7–8** (trees, ensembles, clustering, dimensionality reduction) bridged with **ISLP Ch. 5–6[†]** (resampling, regularisation) + labs.
- **RS-2 Ch. 1–3** (advanced regression, ensembles, model selection) with **ISLP Ch. 8[†]** and **Géron Ch. 6–7**.
- **RS-2 Ch. 4–5** (Bayesian methods, dimensionality reduction in depth) with **ISLP Ch. 12[†]** and **Géron Ch. 8–9**.
- Skim **ISLP Ch. 10[†]** (deep learning intro) — conceptual only.
- **Deliverable:** notebook comparing Lasso, Random Forest, and Gradient Boosting on the same SHARE outcome with cross-validated metrics and SHAP plots.

4. Phase 3 — NLP, classical track (weeks 9–10, 14 – 27 Jul)

Matches the syllabus’s explicit “non-neural” focus.

- **RS-2:** the NLP chapter (text analytics / topic modelling) for the Python-first treatment.
- **Jurafsky & Martin (3e draft)[†]** Ch. 2 (regex/tokenisation), 3 (n-grams), 4 (naive Bayes + sentiment), 5 (logistic regression for text), 6 (vector semantics: TF-IDF, PMI, SVD), 17 (NER), 21 (lexicons/sentiment).
- **NLTK book[†]** Ch. 1, 3, 5, 6, 7 — work the exercises.
- **Bengfort, Bilbro & Ojeda** Ch. 1–5 for sklearn-style text pipelines.
- **Deliverable:** sentiment + TF-IDF classifier on a public corpus (e.g. Reuters), end-to-end pipeline.

5. Phase 4 — Network analysis (weeks 11–12, 28 Jul – 10 Aug)

- **RS-2:** the network-analysis chapter (graphs in Python) as a primer before Menczer.
- **Menczer, Fortunato & Davis** Ch. 1–6 (graphs, centrality, small-world, scale-free, communities) — do the NetworkX notebooks.
- **Barabási[†]** Ch. 2–4, 9 as conceptual companion.
- *Optional:* Newman Ch. 7 + 14 for spectral / community-detection maths.
- **Deliverable:** NetworkX notebook with degree distribution, centrality ranking, and Louvain communities on one real network.

6. Phase 5 — Health economics & SHARE project prep (weeks 13–15, 11 – 31 Aug)

Run in parallel with Phase 4 if time permits.

- **Jones — *Applied Health Economics*:** chapters on health production, insurance demand, and panel methods (~4 chapters).
- **Deb, Norton & Manning** Ch. 1–4 (two-part, count, nonlinear panel) — translate one Stata example to Python (`statsmodels` / `linearmodels`).
- **Cunningham — *The Mixtape*[†]:** refresh DiD and panel FE.
- 3–5 SHARE-specific papers from the Phase 0 shortlist; write a one-paragraph summary of each.
- **easySHARE Release Guide[†]** — full read; mark variables for the project.

- **Deliverable:** 2-page project pre-proposal — research question, SHARE waves/variables, ML/NLP/network angle to satisfy the SDS module, candidate methods, identification strategy.

7. Cross-cutting habits

- One Git repo `sds-prep/` with subfolders `ml/`, `nlp/`, `networks/`, `share/`; each notebook ends with a 5-bullet *what I learned* cell.
- One Zotero / BibTeX file `sds_prep.bib` updated as you read — feeds directly into the semester project.
- Weekly 30-min Friday review: re-skim the week's notes, push commits.

8. Trimming options if time gets tight

- Drop Géron deep-learning chapters (separate spring module).
- Drop Newman; Menczer is enough.
- Replace Bengfort with spaCy tutorials.
- Skip Deb/Norton/Manning if the thesis already covered FE/IV thoroughly.

9. Priority order if you can only do half

RS-1 Ch. 1–6 → ISLP labs (Ch. 2–6, 8) → RS-2 selected chapters → Jurafsky & Martin Ch. 2–6 → Menczer Ch. 1–4 → SHARE methodology + easySHARE guide → Jones (selected chapters).